

2. Ders

$$H = \sum_i \left[\frac{p_i^2}{2m} + \frac{1}{2} m \omega^2 x_i^2 + \frac{k}{2} (x_i - x_{i-1})^2 \right] \quad T+V$$

Legendre Dönüşümü $L = \sum_i p_i \dot{q}_i - H$

$$L = \sum_i \left(m \dot{x}_i \dot{x}_i - \left(\sum_i \frac{m \dot{x}_i^2}{2} + \frac{1}{2} m \omega^2 x_i^2 + \frac{k}{2} (x_i - x_{i-1})^2 \right) \right)$$

$$\Rightarrow \sum_i \left[\frac{1}{2} m \dot{x}_i^2 - \frac{1}{2} m \omega^2 x_i^2 - \frac{k}{2} (x_i - x_{i-1})^2 \right]$$

Süreklilik limitine gidelim;

$$x_i(t) \rightarrow \phi(y,t)$$

$$y \approx a$$

Fark terimi ∂_y dönüşü

Toplamlar integrale

$$\sum_i f(y_i) \approx \frac{1}{a} \int dy f(y)$$

$$\mu = \frac{m}{a} \quad \text{birim uzunluk başına düşen kütle}$$

$$T = k a$$

$$L = \sum_i \frac{1}{2} m \dot{x}_i^2 \Rightarrow L = \frac{1}{2} \int dy \mu (\partial_t \phi)^2$$

$$- \sum_i \frac{k}{2} (x_i - x_{i-1})^2 \Rightarrow L = - \frac{1}{2} \int dy k a (\partial_y \phi)^2 = - \frac{1}{2} \int dy T (\partial_y \phi)^2$$

$$- \sum_i \frac{1}{2} m \omega^2 x_i^2 \Rightarrow - \frac{1}{2} \int dy \mu \omega^2 \phi^2$$

$$L(\phi) = \frac{1}{2} \int dy \mu (\partial_t \phi)^2 - \frac{1}{2} \int dy T (\partial_y \phi)^2 - \frac{1}{2} \int dy \mu \omega^2 \phi^2$$

Kinetik terimin başına $\frac{1}{2}$ gelsin diye $\Phi = \sqrt{\mu} \phi$ tanımlarsak $\tilde{\mu} = T$ $m_{eff} = \omega^2$

$$L(\Phi) = \frac{1}{2} (\partial_t \Phi)^2 - \frac{1}{2} \int dy \tilde{\mu} (\partial_y \phi)^2 - \frac{1}{2} \int dy \mu \omega^2 \phi^2 \quad \text{burda duralım}$$

$\delta S = 0$ koşulunda Klein Gordon denk. üretmeye bakalım,

$$L(q_i, \dot{q}_i) \quad S[q_i(t)] = \int dt L(q_i, \dot{q}_i)$$

$S =$ eylem (action), zaman üzerinden int.

$$\delta S = S[q(t) + \delta q(t)] - S[q] = 0 \quad \text{en küçük } S \text{ gerçek hareket denk çıkarılır.}$$

alan teorisine geçerseniz;

Artık değişkenimiz $q_i(t)$ değil $\phi(y,t)$ dir. (süreklilik alan)

$$\text{yani, } S[\phi(x,t)] = \int dt dy L(\phi, \dot{\phi}, \phi')$$

$$\phi = \phi \quad \phi' = \frac{\partial \phi}{\partial y} \quad \dot{\phi} = \frac{\partial \phi}{\partial t} \quad L = L_{\text{Lagrange}} \text{ jöş.}$$

en yalın haliyle;

$$L = \frac{1}{2} \left[\left(\frac{\partial \phi}{\partial t} \right)^2 - \left(\frac{\partial \phi}{\partial y} \right)^2 - m^2 \phi^2 \right] \quad \text{Bu skaler alan teo'sinin klasik formu}$$

$$\mathcal{L} = \frac{1}{2} \left[\left(\frac{\partial \phi}{\partial t} \right)^2 - \left(\frac{\partial \phi}{\partial y} \right)^2 - m^2 \phi^2 \right] \text{ Bu skalar alan teorisinin klasik formu}$$

$$\delta S = S[q(t) + \delta q(t)] - S[q] = 0$$

$$S[\phi(x,t)] = \int dt dy \mathcal{L}(\phi, \dot{\phi}, \phi')$$

$$\delta S = \int dt dy \mathcal{L}[\phi + \delta\phi, \dot{\phi} + \delta\dot{\phi}, \phi' + \delta\phi'] - \mathcal{L}[\phi, \dot{\phi}, \phi']$$

$$\mathcal{L}(\phi + \delta\phi, \dot{\phi} + \delta\dot{\phi}, \phi' + \delta\phi') \stackrel{(+, -, -)}{\approx} \mathcal{L} + \frac{\partial \mathcal{L}}{\partial \phi} \delta\phi - \frac{\partial \mathcal{L}}{\partial \dot{\phi}} \delta\dot{\phi} - \frac{\partial \mathcal{L}}{\partial \phi'} \delta\phi' \quad \begin{matrix} (?) \\ \int \int \end{matrix} \quad (\text{Taylor açılımı})$$

$$\text{Dolayısıyla } \delta S = \int dt dy \left[\frac{\partial \mathcal{L}}{\partial \phi} \delta\phi - \frac{\partial \mathcal{L}}{\partial \dot{\phi}} \delta\dot{\phi} - \frac{\partial \mathcal{L}}{\partial \phi'} \delta\phi' \right]$$

$$\delta\phi = \delta\phi$$

$$\delta\dot{\phi} = \delta \frac{d}{dt} \phi = \frac{d}{dt} \delta\phi$$

* varyasyon ve türev işlemleri birbirini yerine gelebilir.

$$\delta\phi' = \delta \frac{d}{dy} \phi = \frac{d}{dy} \delta\phi$$

$$\Rightarrow \delta S = \int dt dy \left[\frac{\partial \mathcal{L}}{\partial \phi} \delta\phi - \frac{\partial \mathcal{L}}{\partial \dot{\phi}} \frac{d}{dt} \delta\phi - \frac{\partial \mathcal{L}}{\partial \phi'} \frac{d}{dy} \delta\phi \right]$$

$$\Rightarrow \int dt dy \left[\frac{\partial \mathcal{L}}{\partial \phi} - \frac{\partial \mathcal{L}}{\partial \dot{\phi}} \frac{d}{dt} - \frac{\partial \mathcal{L}}{\partial \phi'} \frac{d}{dy} \right] \delta\phi = 0$$

$$\Rightarrow \int dt dy \underbrace{\left[\frac{\partial \mathcal{L}}{\partial \phi} - \frac{d}{dt} \frac{\partial \mathcal{L}}{\partial \dot{\phi}} - \frac{d}{dy} \frac{\partial \mathcal{L}}{\partial \phi'} \right]}_0 \delta\phi = 0$$

$$T = \mu v^2$$

$$\Rightarrow \frac{\partial \mathcal{L}}{\partial \phi} - \frac{d}{dt} \frac{\partial \mathcal{L}}{\partial \dot{\phi}} - \frac{d}{dy} \frac{\partial \mathcal{L}}{\partial \phi'} = 0 \quad \mathcal{L} = \int dy \left[\frac{1}{2} m \dot{x}_i^2 - \frac{1}{2} m \omega^2 x_i^2 - \frac{1}{2} k (x_i - x_{i-1})^2 \right]$$

$$\mathcal{L}(\phi) = \int dy \mu (d_t \phi)^2 - \frac{1}{2} T (d_y \phi)^2 - \frac{1}{2} \mu \omega^2 \phi^2$$

$$\Rightarrow \mathcal{L}(\phi) = \int dy \mu \left[\frac{1}{2} (d_t \phi)^2 - \frac{1}{2} v^2 (d_y \phi)^2 - \frac{1}{2} \omega^2 \phi^2 \right]$$

$$\mathcal{L}(\phi) = \int dy \mu \left\{ \frac{1}{2} (d_t \phi)^2 - \frac{1}{2} v^2 (d_y \phi)^2 - \frac{1}{2} \omega^2 \phi^2 \right\} \quad \frac{1}{\omega^2} \text{ aşağıda istiyoruz}$$

$$\mathcal{L}(\phi) = \int dy \mu \left\{ \frac{1}{2 \omega^2} (d_t \phi)^2 - \frac{1}{2} (d_y \phi)^2 - \frac{1}{2} m^2 \phi^2 \right\} \quad m \approx \frac{\omega}{v} \quad 1 \leq k=1$$

$$\Rightarrow \int dy \left[\frac{1}{2} (d_t \phi)^2 - (d_y \phi)^2 - \frac{1}{2} m^2 \phi^2 \right]$$

$$\int dy \left[\frac{1}{2} v^2 (\dot{\phi})^2 - \frac{1}{2} (\phi')^2 - \frac{1}{2} m^2 \phi^2 \right]$$

$$\delta \dot{\phi} = \frac{d}{dt} \delta \phi \quad \delta \phi' = \frac{d}{dy} \delta \phi \quad \dot{\phi} = \frac{d\phi}{dt} \quad \phi' = \frac{d\phi}{dy}$$

$$\frac{d\mathcal{L}}{d\phi} - \frac{d}{dt} \frac{d\mathcal{L}}{d\dot{\phi}} - \frac{d}{dy} \frac{d\mathcal{L}}{d\phi'} = 0$$

$$\frac{d\mathcal{L}}{d\phi} = -m^2 \phi \quad \frac{d\mathcal{L}}{d\dot{\phi}} = \frac{1}{v^2} \dot{\phi} \quad \frac{d\mathcal{L}}{d\phi'} = -\phi'$$

$$-m^2 \phi - \frac{1}{v^2} \frac{d^2}{dt^2} \phi + \frac{d^2}{dy^2} \phi = 0$$